SHUBHAM MOHAPATRA

AI-Native Software Engineer & Full-Stack Builder

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SUMMARY

Computer Science engineer (B.Tech) building AI systems that act — autonomous agents with function calling — and the production-grade full-stack products around them. Hackathon winner (DRDO co-partnered hackathon) with a fast-shipper mindset: idea to production.

EXPERIENCE

SentrovaSolutions AI-Native Engineer — Present

- Design and build agentic AI workflows — tool-using agents that plan, call functions and complete multi-step tasks end to end.
- Full-stack development across React / Next.js frontends and Node.js / FastAPI services backed by PostgreSQL and pgvector.
- Ship production-grade solutions: from prototype to deployed, monitored releases.

Hansa AI / Hamaraneta Software Development Intern — Internship

- Engineering research to evaluate approaches and tools before they went into the product.
- Integrated a chatbot into the platform, connecting conversational flows to backend APIs.
- Built data collection pipelines that gathered and structured data for downstream use.

FEATURED PROJECTS

Agentic Workflow Engine
Python · FastAPI · Function Calling · PostgreSQL · Redis · React

- Agent loop using function calling: plan → call tool → observe → decide next step until the goal is met.
- Schema-validated tool registry; run state persisted in PostgreSQL for resumable, auditable runs.

OCR → JSON Pipeline
Python · FastAPI · OCR · LLM extraction · PostgreSQL · Docker

- OCR data pipeline converting scanned pages and images into structured, schema-validated JSON.

Vector Search Assistant
Next.js · TypeScript · Node.js · PostgreSQL · pgvector

- Semantic search and retrieval-augmented answers with sources, using pgvector inside PostgreSQL.

TECHNICAL SKILLS

Languages:	Python, JavaScript, TypeScript, Java, C/C++
Frontend:	React, Next.js, Tailwind CSS, HTML5/CSS3
Backend & Systems:	Node.js, Express, FastAPI, REST APIs
Databases & Vector Stores:	PostgreSQL, pgvector, Redis, ClickHouse
AI & Tooling:	Autonomous Agents, Function Calling, Git/GitHub, Docker, Linux/Bash

ACHIEVEMENTS

- Winner — DRDO co-partnered hackathon.
- Competed at national-level hackathons and engineering events.

EDUCATION

B.Tech — Computer Science & Engineering